

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element G Initial Template

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation (G1)

Ensure that photos are taken from multiple angles so that all sides and subsystems can be seen. All images should have captions and should be referenced in-text.

Prototype Construction of Design #1 (Cardboard)

Date: 13/5/2024

Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:

No photos available

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	Cut up 6 pieces of cardboard into 16 in x 3 in rectangle, fold into rectangular prism	X-Acto knife, hot glue gun, folding cardboard
Step 2	Cut 28 in x 15 in cardboard piece	X-Acto knife, cutting cardboard
Step 3	Glue 6 triangular prisms equally spaced in two rows onto the the piece cut out in step 2	Hot glue gun, gluing cardboard

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Step 4		
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Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

We were planning to do a full recreation of the shelving system in the blueprint, but throughout the construction we realized that we could test all the design requirements while just building a smaller section of it, and since the design is modular, testing that part would work just as well as testing a full prototype shelving unit.

Final Prototype Construction of Design #1 (Cardboard)

Date: 22/5/2024

Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:

No photos available

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	Cut up 6 pieces of cardboard into 16 in x 3 in rectangle, fold into rectangular prism	X-Acto knife, hot glue gun, folding cardboard
Step 2	Cut 28 in x 15 in cardboard piece	X-Acto knife, cutting cardboard
Step 3	Glue 6 triangular prisms equally spaced in two rows onto the the piece cut out in step 2	Hot glue gun, gluing cardboard

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Step 4	Add extra glue to joints for stability	Hot glue gun, gluing cardboard
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Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

See above. We were planning to do a full recreation of the shelving system in the blueprint, but throughout the construction we realized that we could test all the design requirements while just building a smaller section of it, and since the design is modular, testing that part would work just as well as testing a full prototype shelving unit.

From before, we also decided to add more glue to make it more stable.

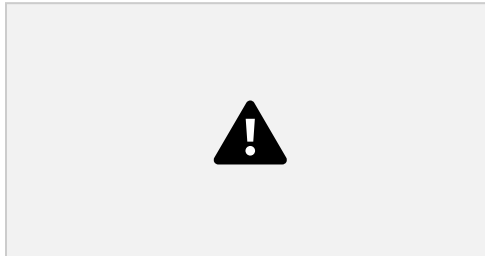
Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
Cardboard	~4 sq ft.	Obtained in classroom	\$0	2	\$0
X-Acto Knife	N/A	Obtained in classroom	\$0	3	\$0
Hot glue gun	N/A	Obtained in classroom	\$0	1	\$0

Detailed sketches, which might include CAD if you have learned it in your class:

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Construction of the Final Prototype #1 (cardboard) (G2)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
The design must be able to store up to 200 rolls of 3d printer filament.	The 48 dowels on the shelving are the same length and can hold the same number of 3D filaments. If they could each hold 5 filament rolls, $48 * 5 = 240 > 200$ dowels, so it would be adequate.
Must be able to support up to 200 rolls weighing 1 kg each without breaking or sinking.	We could not test this on cardboard. Expert review needed.
Rolls must be organized by their specific color of filament, size of rolls, thickness of filament, and material of filament.	We can test this by organizing rolls on the dowels. If we are able to organize them by each of these features, the prototype is adequate. Otherwise, it isn't.
The design needs to store the rolls out of the way and	We can bring it to Mr. L's room and put it in place to see if it sticks out or interferes with anything. We could also ask Mr. L if it interferes with anything.

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without interfering with any other important places/things in the room.	
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Testing or Modeling of Attributes (*G3, G4*) and Addressing Non-Testable Attributes (*G4*)

Describe all of your prototype's attributes (features, aspects, details, or subsystems). Demonstrate that your prototype is constructed with sufficient functionality that your prototype can test all attributes of your design. This requires you to look back at your initial proposed design details from Element D.

Attribute	Description	Describe your prototype's functionality to test this attribute
Rows of shelving	Rows with dowels	The prototype has these rows of shelving with dowels
Dowels	N/A	The prototype has dowels that can hold 3D printer filament
Holds 3D printer filament	The dowels can store 3D printer filament	The prototype holds 3D printer filament
Holds up on shelf	Can support itself on Mr. L's steel shelf	The prototype is able to support itself on Mr. L's steel shelf



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Addressing Non-Testable Attributes (G4)

For all attributes that you listed "expert review required" in the table above, provide a justification for why that is true. "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

Attribute	Justification for why this attribute cannot be modeled or tested with your prototype
Must be able to support up to 200 rolls weighing 1 kg each without breaking or sinking.	Because the cardboard cannot be expected to hold up the full 200kg of filament rolls, only the later wood design would have that weight-bearing capacity. We would need a field expert to do a very detailed model of the forces involved and test the weight-bearing capacity of the actual wood we have, and then evaluate its ability.

Construction of the Final Prototype #2 (wood) (G2)

We were unable to complete a final prototype for design 2 due to time constraints. It would've been a more complete prototype envisioned earlier, built of wood and to full dimensions.

Photos:



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What the top shelf was envisioned to look like by the end. The holes were for PVC dowels, the black 3D printed part was the PVC connector.



Closeup of the PVC connector, with threading to work like a screw, with one end on the PVC pipe, the other end in the wood.

Design Requirement	How will this prototype allow you to collect objective, measurable data?
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The design must be able to store up to 200 rolls of 3d printer filament.	N/A
Must be able to support up to 200 rolls weighing 1 kg each without breaking or sinking.	N/A
Rolls must be organized by their specific color of filament, size of rolls, thickness of filament, and material of filament.	N/A
The design needs to store the rolls out of the way and without interfering with any other important places/things in the room.	N/A

Testing or Modeling of Attributes (*G3*, *G4*) and Addressing Non-Testable Attributes (*G4*)

Name: _____

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Attribute	Description	Describe your prototype's functionality to test this attribute
Rows of shelving	Rows with dowels	N/A
Dowels	N/A	N/A
Holds 3D printer filament	The dowels can store 3D printer filament	N/A
Holds up on shelf	Can support itself on Mr. L's steel shelf	N/A

Addressing Non-Testable Attributes (G4)

For all attributes that you listed "expert review required" in the table above, provide a justification for why that is true. "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

Attribute	Justification for why this attribute cannot be modeled or tested with your prototype
There are no non-testable attributes	N/A

